



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 01

1. **Title:** Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.

2. **Software/Tools Required:** HTML, VS Code.

3. **Theory:** HTML (HyperText Markup Language) is the standard language used to create web pages. It provides the basic structure of a webpage using elements such as headings, paragraphs, images, lists, tables, and buttons. CSS (Cascading Style Sheets) is used to style the webpage. It controls the layout, colors, fonts, spacing, and overall appearance of the webpage. In this program, an "About Me" webpage is created using HTML and CSS. The webpage contains a profile picture, personal information, skills, education details, and a contact button. CSS is used to improve the appearance by adding colors, spacing, borders, and alignment.

4. Output:

← ↻ file:///D:/Hp/Documents/JSP1.html1189

Profile Picture

Rutva Sawarkar

B.Tech Computer Science Student

About Me

Hello! My name is Rutva Sawarkar. I am pursuing B.Tech in Computer Science. I enjoy learning new technologies and improving my programming skills. My areas of interest include Web Development, Python, Artificial Intelligence, and Data Science.

Skills

- HTML & CSS
- Python
- Java
- SQL
- Problem Solving

Education

Course	Institute	Year
B.Tech CSE	Symbiosis Institute of Technology	2024 - Present

Contact Me



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Case Study No.: 01

Case Study Title: Create a webpage for college department and link the student information page with it. Write a JavaScript program to display “Welcome to SIT nagpur” in an alert box.

Case Study Program Code:

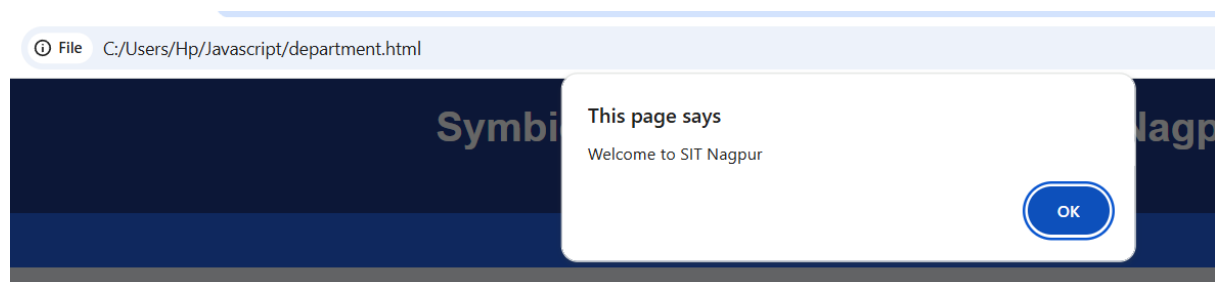
```
<> department.html > html > body > div.container > div.card > p
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  <title>SIT Nagpur - Computer Science Department</title>
7
8  <style>
9  *{
10     margin:0;
11     padding:0;
12     box-sizing:border-box;
13     font-family:Arial, sans-serif;
14 }
15
16 body{
17     background: #f4f7fb;
18 }
19
20 header{
21     background: #1e3a8a;
22     color: white;
23     text-align:center;
24     padding:20px;
25 }
26
27 nav{
28     background: #2563eb;
29     text-align:center;
30     padding:12px;
31 }
32
33 nav a{
34     color: white;
35     text-decoration:none;
36     margin:0 15px;
37     font-weight:bold;
```



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```
< student.html > < html > < body > < div.container > < table > < tr > < td
2  <html lang="en">
70 <body>
72 <header>
73   <h1>Student Information</h1>
74 </header>
75
76 <div class="container">
77
78 <table>
79 <tr>
80   <th>Roll No</th>
81   <th>Name</th>
82   <th>Branch</th>
83   <th>Year</th>
84 </tr>
85
86 <tr>
87   <td>101</td>
88   <td>Rahul Sharma</td>
89   <td>CSE</td>
90   <td>Second Year</td>
91 </tr>
92
93 <tr>
94   <td>102</td>
95   <td>Priya Verma</td>
96   <td>CSE</td>
97   <td>Second Year</td>
98 </tr>
99
100 <tr>
101   <td>103</td>
102   <td>Aman Patil</td>
103   <td>CSE</td>
104   <td>Second Year</td>
105 </tr>
106
```

OUTPUT:





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SIT Nagpur - Computer Science & Engineering Department

Symbiosis Institute of Technology, Nagpur

Computer Science & Engineering Department

[Home](#) [Student Information](#)

About the Department

The Computer Science & Engineering Department offers quality education in programming, artificial intelligence, data science, cybersecurity, cloud computing, and software engineering.

Students are encouraged to participate in research, internships, hackathons, and technical workshops for overall development.

[View Student Information](#)

© 2026 SIT Nagpur | Department of Computer Science & Engineering

Student Information

Student Information

Roll No	Name	Branch	Year
101	Rahul Sharma	CSE	Second Year
102	Priya Verma	CSE	Second Year
103	Aman Patil	CSE	Second Year

[Back to Department](#)

Conclusion: The "About Me" webpage was successfully created using HTML and CSS. The webpage displays personal information in a well-organized and visually appealing format. This practical demonstrates the basic use of HTML for webpage structure and layout.



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Experiment No.: 02

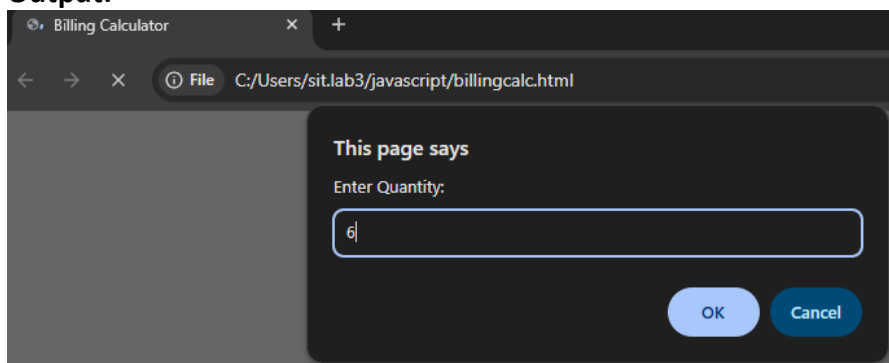
1. Title: Use var, let, const, and demonstrate basic features like template literals and destructuring, build a billing calculator with user input.

2. Software/Tools Required: HTML, VS Code.

3. Theory: JavaScript is a scripting language used to make web pages interactive and dynamic. It allows developers to perform calculations, validate user input, display messages, and manipulate webpage content. In this experiment, a **Billing Calculator** is developed using JavaScript. The program accepts user inputs such as customer name, product name, quantity, and price per item through dialog boxes (prompt()). It then calculates the total bill and displays a formatted billing receipt on the webpage. The program demonstrates the following JavaScript features:

- **var** – Used to declare function-scoped variables. It can be redeclared and updated.
- **let** – Used to declare block-scoped variables. It can be updated but not redeclared within the same scope.
- **const** – Used to declare constants whose values cannot be reassigned.
- **Template Literals** – Enclosed within backticks (`), they allow embedding variables directly into strings using `\${}` for easier string formatting.
- **Destructuring** – A feature that extracts values from arrays or objects into separate variables, making the code cleaner and more readable.

4. Output:





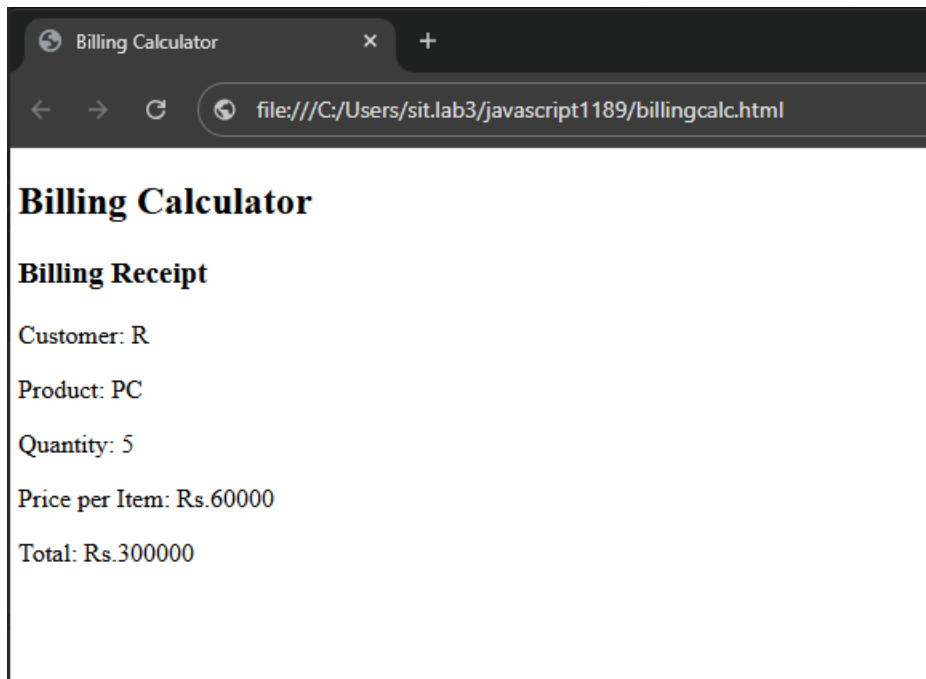
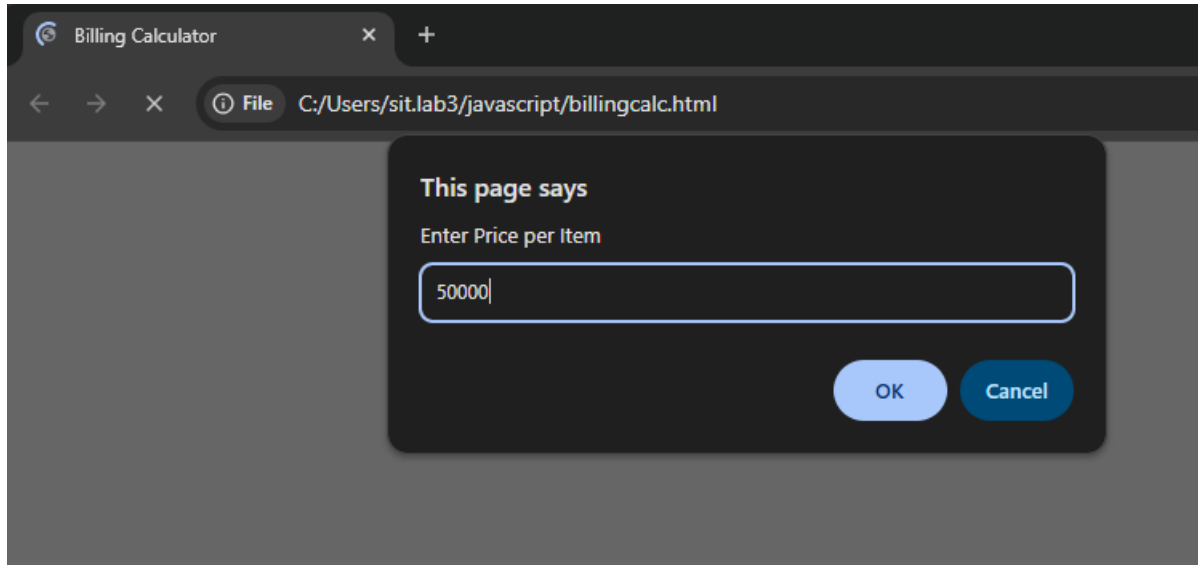
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CASE STUDY 02

Case Study Title: Develop a webpage for an online shopping website using HTML, CSS, and HTML5 tags, and calculate the bill.

Code:

```
Welcome Practical03.html pract04.html department.html
shopping.html > html
2 <html lang="en">
99 <body>
112 <section>
139 </section>
140
141 <footer>
142 © 2026 ABC Online Shopping
143 </footer>
144
145 <script>
146
147 function calculateBill(){
148
149 let laptop=Number(document.getElementById("laptop").value);
150 let headphone=Number(document.getElementById("headphone").value);
151 let mouse=Number(document.getElementById("mouse").value);
152
153 let total=(laptop*50000)+(headphone*2000)+(mouse*700);
154
155 document.getElementById("bill").innerHTML=
156 "<h2>Bill Summary</h2>"+
157 "<p>Laptop: ₹"+(laptop*50000)+"</p>"+
158 "<p>Headphones: ₹"+(headphone*2000)+"</p>"+
159 "<p>Mouse: ₹"+(mouse*700)+"</p>"+
160 "<hr>"+
161 "<h2>Total Bill = ₹"+total+"</h2>";
162
163 }
164
165 </script>
166
167 </body>
168 </html>
```



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Output:

The screenshot displays a web browser window with the title 'Online Shopping Website'. The address bar shows the file path: file:///C:/Users/Hp/Javascript/shopping.html1189Rutva. The website header is 'ABC Online Shopping' with the tagline 'Shop Smart • Shop Easy' and navigation links for 'Home', 'Products', and 'Contact'. The main content area features a shopping cart with three items: a Laptop (Price: ₹50000, Quantity: 1), Headphones (Price: ₹2000, Quantity: 2), and a Mouse (Price: ₹700, Quantity: 1). A green 'Calculate Bill' button is located below the cart items. The 'Bill Summary' section shows the following details: Laptop: ₹50000, Headphones: ₹4000, Mouse: ₹700, and a bolded 'Total Bill = ₹54700'. The footer of the website states '© 2026 ABC Online Shopping'.

Conclusion:

The Online Shopping Website successfully demonstrates the use of HTML, CSS, HTML5 semantic tags, and JavaScript to create a simple and interactive e-commerce webpage. It enables users to select product quantities and automatically calculates the total bill, reducing manual effort and errors. The project highlights the use of webpage structuring, styling, user input handling, event-driven programming, and dynamic bill generation, making it an effective example of a basic online shopping application.



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Experiment No.: 03

- Title:** Implement control structures and form validation; create a grading system based on user-entered marks.
- Software/Tools Required:** HTML, JavaScript, CSS, VS Code.
- Theory:** This program implements a **Student Grading System** using **HTML, CSS, and JavaScript**. It accepts the student's name and marks for five subjects through a form. JavaScript performs **form validation** to ensure that all fields are filled and that the marks entered are within the valid range (0–100). The program then calculates the **total marks, percentage, grade, and pass/fail status** using **control structures** (if, else if, else) and **loops** (for). The result is displayed dynamically on the webpage without reloading it, providing an interactive user experience.

4. Output:

Grading System

file:///C:/Users/Hp/Javascript/1189Practical03.html

Rutva

Student Grading System

76

Submit

Marks: 76
Grade: B



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CASE STUDY 03

Case Study Title: Scholarship Eligibility Checker Using JavaScript

Code:

```
Welcome Practical03.html pract04.html department.html billing.html index.html
grades.html > html
2 <html lang="en">
3 <head>
8 <style>
66 #result{
67     margin-top:20px;
68     padding:15px;
69     background: #f4f4f4;
70     border-radius:8px;
71     text-align:center;
72     font-size:18px;
73 }
74 </style>
75
76 </head>
77 <body>
78
79 <div class="container">
80
81 <h2>Scholarship Eligibility</h2>
82
83 <label>Enter Marks (0-100)</label>
84
85 <input type="number" id="marks" placeholder="Enter Marks">
86
87 <button onclick="checkScholarship()">Check Eligibility</button>
88
89 <div id="result"></div>
90
91 </div>
92
93 <script>
94
95 function checkScholarship(){
96
97     let marks=Number(document.getElementById("marks").value);
98
99     let scholarship="";
100
101     if(marks<0 || marks>100){
102         scholarship="Please enter valid marks.";
103     }
104     else if(marks<50){
105         scholarship="Not Eligible";
106     }
107     else if(marks<=69){
108         scholarship="Bronze Scholarship";
109     }
110     else if(marks<=84){
111         scholarship="Silver Scholarship";
112     }
113     else if(marks<=94){
114         scholarship="Gold Scholarship";
115     }
116     else{
117         scholarship="Platinum Scholarship";
118     }
119
120     document.getElementById("result").innerHTML=
121     "<b>Result:</b><br>"+scholarship;
122
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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Output:

Screenshot of a web browser showing a "Scholarship Eligibility" form. The browser tab is labeled "Scholarship Eligibility" and the address bar shows a file path. The form is titled "Scholarship Eligibility" and asks the user to "Enter Marks (0-100)". The input field contains the value "89". A blue button labeled "Check Eligibility" is present. Below the button, the result is displayed as "Result: Gold Scholarship".

Result/Conclusion: The Student Grading System successfully demonstrates the use of **HTML for structure**, **CSS for styling**, and **JavaScript for logic and validation**. It applies control structures, loops, arrays, and DOM manipulation to process user input and generate results. This project improves the accuracy of grade calculation, reduces manual errors, and provides immediate feedback, making it a simple and effective web-based application for evaluating student performance.



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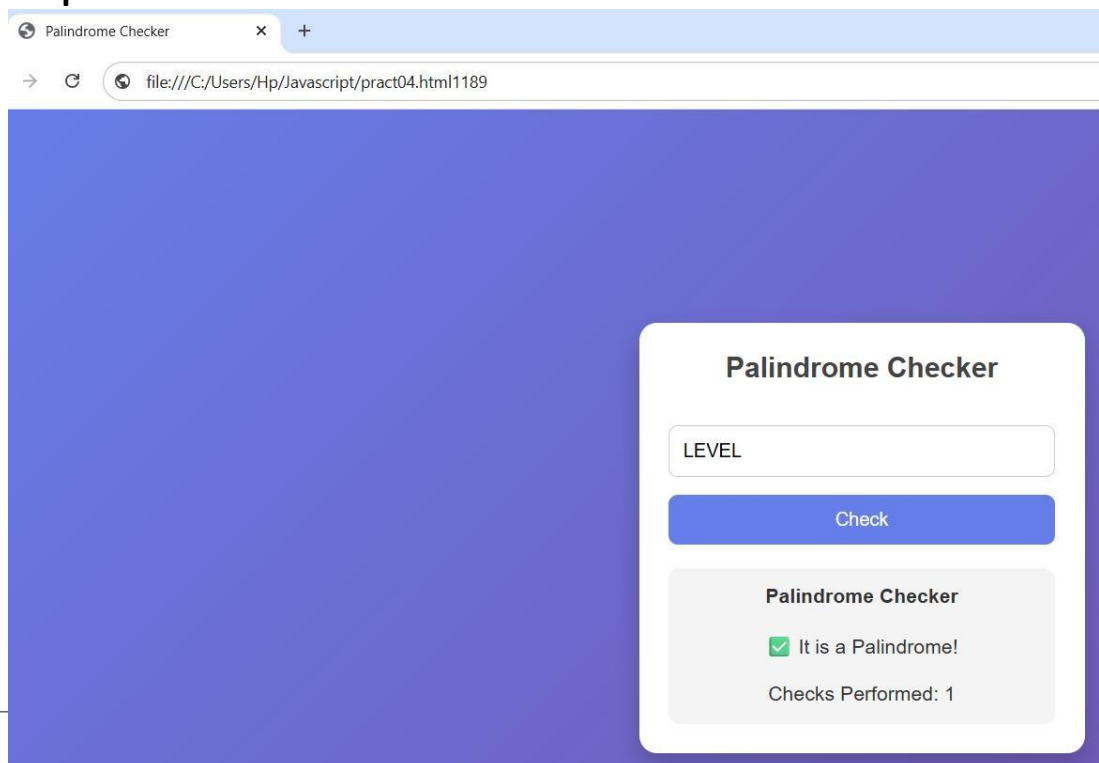
Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

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Experiment No.: 04

- 1. Title:** Use function types, scope, and closures; apply try-catch; build a palindrome checker.
- 2. Software/Tools Required:** HTML, JavaScript, CSS, VS Code.
- 3. Theory:** This program implements a Palindrome Checker using HTML, CSS, and JavaScript. It allows the user to enter a word or sentence and checks whether it is a palindrome. The program demonstrates different types of functions, including a function declaration, function expression, and arrow function. It also illustrates variable scope by using both global and local variables. A closure is implemented to count the number of palindrome checks performed while preserving the counter value between function calls. Exception handling is achieved using try...catch, which displays an appropriate error message if the input is empty. The webpage dynamically displays the result without refreshing the page.

4. Output:





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Palindrome Checker

×

+

→ ↺ file:///C:/Users/Hp/Javascript/pract04.html1189

Palindrome Checker

Check

Palindrome Checker
✗ It is NOT a Palindrome.
Checks Performed: 2



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CASE STUDY 04

Case Study Title: To implement an ATM Card PIN Verification system using JavaScript by demonstrating function declaration, function expression, arrow functions, variable scope, closures, and exception handling.

Code:

```
Welcome Practical03.html pract04.html department.html billing.html index.html
ATM.html > html
2 <html lang="en">
73 <body>

87 <script>
88
89 // Global Scope
90 let storedPIN = "1234";
91
92 // Function Declaration
93 function createCounter(){
94     let attempts = 0;
95
96     // Closure
97     return function(){
98         attempts++;
99         return attempts;
100     };
101 }
102
103 const counter = createCounter();
104
105 // Function Expression
106 const isValidPIN = function(pin){
107     return /^\d{4}$/.test(pin);
108 };
109
110 // Arrow Function
111 const checkPIN = pin => pin === storedPIN;
112
113 // Main Function
114 function verifyPIN(){
115
116     try{
117
118         let pin = document.getElementById("pin").value;
119
120         if(pin=="")
121             throw "Please enter your PIN.";
122
123         if(!isValidPIN(pin))
124             throw "PIN must contain exactly 4 digits.";
125
126         let message;
127
128         if(checkPIN(pin))
129             message = "✅ PIN Verified Successfully!";
130         else
131             message = "❌ Incorrect PIN.";
132
133         document.getElementById("result").innerHTML =
134             message +
135             "<br><br>Verification Attempts: " + counter();
136
137     }
138
139     catch(error){
140
141         document.getElementById("result").innerHTML =
142             "<span style='color:red'>" + error + "</span>";
143
144     }
145 }
```



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Output:

The screenshot shows a web browser window with a single tab titled 'ATM PIN Verification'. The address bar displays the file path: `file:///C:/Users/Hp/Javascript/ATM.html1189Rutva`. The main content area has a dark blue background. On the right side, there is a white card with the title 'ATM PIN Verification'. Inside the card, there is a text input field containing four dots (****). Below the input field is a dark blue button labeled 'Verify PIN'. At the bottom of the card, there is a light gray box containing a red 'X' icon, the text 'Incorrect PIN.', and 'Verification Attempts: 2'.

Result/Conclusion: The Palindrome Checker successfully demonstrates important JavaScript concepts such as function types, variable scope, closures, and exception handling (try...catch). It validates user input, processes the entered text by removing spaces and special characters, and determines whether it is a palindrome. The project provides a user-friendly interface and showcases how JavaScript can be used to build interactive, reliable, and efficient web applications.

Experiment No.: 5

- 1. Title:** Use string functions and regex for email validation, data extraction, and text analysis.
- 2. Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. Theory: String functions and Regular Expressions (Regex) are used in JavaScript to process and analyze text. String functions such as `toLowerCase()`, `toUpperCase()`, `includes()`, `split()`, `trim()`, and `replace()` help manipulate and search text. Regular Expressions are patterns used to find specific text formats, such as validating an email address, extracting numbers or words, and identifying patterns in a sentence. In a basic HTML, CSS, and JavaScript application, HTML provides the input fields and structure, CSS provides styling, and JavaScript uses string functions and Regex to validate emails, extract required data, and perform simple text analysis.

4. Output :

127.0.0.1:5500/pract06.html1189Rutva

Email Validation, Data Extraction & Text Analysis

Enter your text:

```
Hello, my name is Rutva Sawarkar.  
My email addresses are rutva@gmail.com and rutva123@yahoo.com.  
You can contact me at 9876543210.  
I am a B.Tech CSE student learning Python, Data Analysis and Machine Learning.  
Python is easy to learn and Python is very useful for data analysis.  
Regular expressions are useful for email validation and text processing.
```

Analyze Text

1. Email Validation

rutva@gmail.com → **Valid**
rutva123@yahoo.com → **Valid**

2. Phone Number Extraction

9876543210

3. Text Analysis

Total Characters: **351**
Total Words: **55**
Unique Words: **40**

4. Word Frequency

Frequency of "python": 3

5. Words Starting With 'P'

python python python processing

6. Words Ending With 'ING'

learning learning processing

7. Words Longer Than 6 Characterssawarkar addresses rutvagmailcom rutva123yahoom contact 9876543210 student learning
analysis machine learning analysis regular expressions validation processing**8. String Functions****Uppercase Text:**

HELLO, MY NAME IS RUTVA SAWARKAR. MY EMAIL ADDRESSES ARE RUTVA@GMAIL.COM AND RUTVA123@YAHOO.COM. YOU CAN CONTACT ME AT 9876543210. I AM A B.TECH CSE STUDENT LEARNING PYTHON, DATAANALYSIS AND MACHINE LEARNING. PYTHON IS EASY TO LEARN AND PYTHON IS VERY USEFUL FOR DATA ANALYSIS. REGULAR EXPRESSIONS ARE USEFUL FOR EMAIL VALIDATION AND TEXT PROCESSING.

Lowercase Text:

hello, my name is rutva sawarkar. my email addresses are rutva@gmail.com and rutva123@yahoo.com. you can contact me at 9876543210. i am a b.tech cse student learning python, data analysis and machine learning. python is easy to learn and python is very useful for data analysis. regular expressions are useful for email validation and text processing.

Text starts with "Hello": **true**Text contains "Python": **true**

Number of occurrences of "Python": 3

Conclusions: String functions and regular expressions (Regex) are important features of JavaScript that make text processing easier and more efficient. String functions can be used to search, compare, split, replace, trim, and change text, while Regex helps identify and work with specific patterns in the given data. These techniques can be applied to email validation, data extraction, keyword searching, and basic text analysis. When combined with HTML for creating the webpage and CSS for styling, JavaScript provides the required functionality to make the application interactive and user-friendly. Overall, learning string functions and Regex provides a strong foundation for handling text-based data and developing practical web applications.



Case Study No.: 5

1. Case Study Title:

To study the functioning of Arrays and Loops in JavaScript

- Create and manipulate arrays using push(), pop(), shift(), unshift(), splice(), and slice().
- Use array methods map(), filter(), reduce(), and forEach().

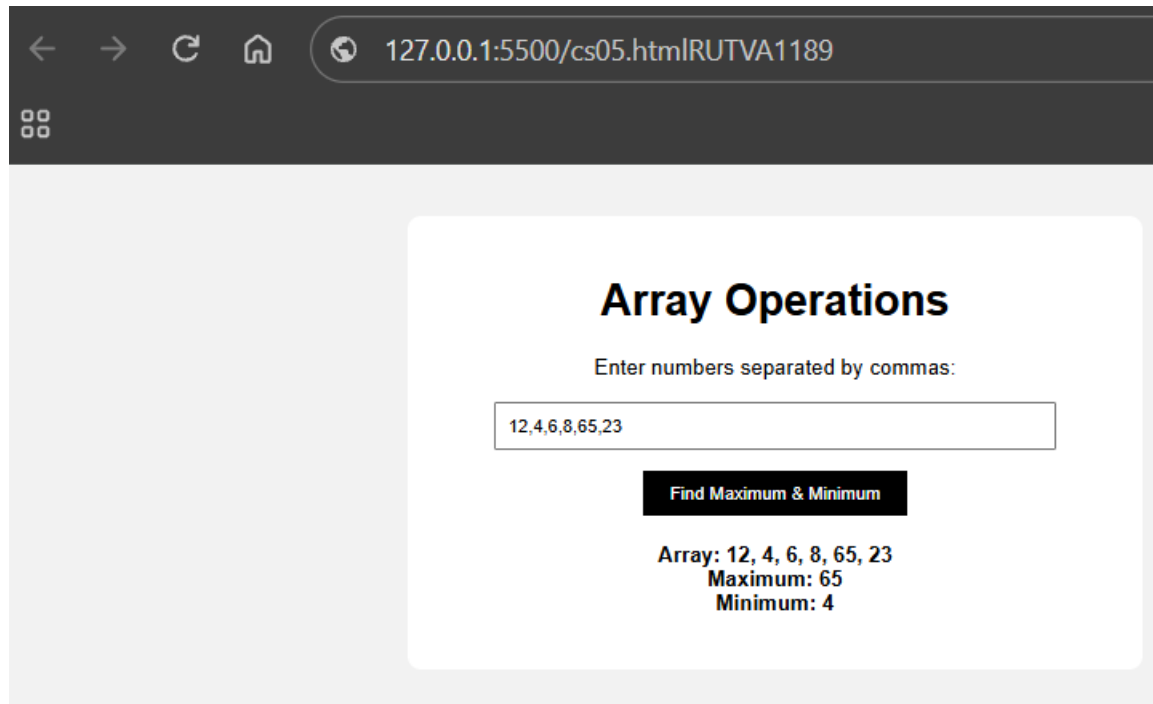
Case Study: Create an array of numbers and find the maximum and minimum values.

2. Case Study Program Code:

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Array Operations</title>
5
6      <style>
7          body {
8              font-family: Arial;
9              background-color: #f2f2f2;
10             text-align: center;
11             padding: 30px;
12         }
13
14         .box {
15             background: white;
16             width: 500px;
17             margin: auto;
18             padding: 25px;
19             border-radius: 10px;
20         }
21
22         input {
23             padding: 10px;
24             width: 80%;
25         }
26
27         button {
28             padding: 10px 20px;
29             margin-top: 15px;
30             background-color: black;
31             color: white;
32             border: none;
33         }
34
35         #result {
36             margin-top: 20px;
37             font-weight: bold;
38         }
39     </style>
40 </head>
41
42 <body>
```

```
43
44 <div class="box">
45     <h1>Array Operations</h1>
46
47     <p>Enter numbers separated by commas:</p>
48
49     <input type="text" id="numbers"
50         |   placeholder="10, 20, 5, 40, 15">
51
52     <br>
53
54     <button onclick="calculate()">Find Maximum & Minimum</button>
55
56     <p id="result"></p>
57 </div>
58
59 <script>
60
61 function calculate() {
62
63     let input = document.getElementById("numbers").value;
64
65     // Convert input into array
66     let arr = input.split(",").map(Number);
67
68     // Find maximum and minimum
69     let max = Math.max(...arr);
70     let min = Math.min(...arr);
71
72     document.getElementById("result").innerHTML =
73         "Array: " + arr.join(", ") +
74         "<br>Maximum: " + max +
75         "<br>Minimum: " + min;
76 }
77
78 </script>
79
80 </body>
81
82 </html>
```

3. Output:



The screenshot shows a web browser window with the address bar displaying '127.0.0.1:5500/cs05.htmlRUTVA1189'. The page content is titled 'Array Operations' and includes a prompt 'Enter numbers separated by commas:'. A text input field contains the string '12,4,6,8,65,23'. Below the input is a black button with the text 'Find Maximum & Minimum'. The output of the operation is displayed as 'Array: 12, 4, 6, 8, 65, 23', 'Maximum: 65', and 'Minimum: 4'.

4. Result/Conclusion:

Thus, the functioning of JavaScript arrays and loops was studied and implemented successfully. Various array methods such as `push()`, `pop()`, `shift()`, `unshift()`, `splice()`, `slice()`, `map()`, `filter()`, `reduce()`, and `forEach()` were used to manipulate and process array elements. The maximum and minimum values were also successfully found from an array of numbers.

Experiment No.: 6

- 1. Title:** Use string functions and regex for email validation, data extraction, and text analysis.
- 2. Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. Theory: String functions and Regular Expressions (Regex) are used in JavaScript to process and analyze text. String functions such as `toLowerCase()`, `toUpperCase()`, `includes()`, `split()`, `trim()`, and `replace()` help manipulate and search text. Regular Expressions are patterns used to find specific text formats, such as validating an email address, extracting numbers or words, and identifying patterns in a sentence. In a basic HTML, CSS, and JavaScript application, HTML provides the input fields and structure, CSS provides styling, and JavaScript uses string functions and Regex to validate emails, extract required data, and perform simple text analysis.

4. Output :

127.0.0.1:5500/pract06.html1189Rutva

Email Validation, Data Extraction & Text Analysis

Enter your text:

```
Hello, my name is Rutva Sawarkar.  
My email addresses are rutva@gmail.com and rutva123@yahoo.com.  
You can contact me at 9876543210.  
I am a B.Tech CSE student learning Python, Data Analysis and Machine Learning.  
Python is easy to learn and Python is very useful for data analysis.  
Regular expressions are useful for email validation and text processing.
```

Analyze Text

1. Email Validation

rutva@gmail.com → **Valid**
rutva123@yahoo.com → **Valid**

2. Phone Number Extraction

9876543210

3. Text Analysis

Total Characters: **351**
Total Words: **55**
Unique Words: **40**

4. Word Frequency

Frequency of "python": 3

5. Words Starting With 'P'

python python python processing

6. Words Ending With 'ING'

learning learning processing

7. Words Longer Than 6 Characters

sawarkar addresses rutvagmailcom rutva123yahoom contact 9876543210 student learning
analysis machine learning analysis regular expressions validation processing

8. String Functions

Uppercase Text:

HELLO, MY NAME IS RUTVA SAWARKAR. MY EMAIL ADDRESSES ARE RUTVA@GMAIL.COM AND RUTVA123@YAHOO.COM. YOU CAN CONTACT ME AT 9876543210. I AM A B.TECH CSE STUDENT LEARNING PYTHON, DATAANALYSIS AND MACHINE LEARNING. PYTHON IS EASY TO LEARN AND PYTHON IS VERY USEFUL FOR DATA ANALYSIS. REGULAR EXPRESSIONS ARE USEFUL FOR EMAIL VALIDATION AND TEXT PROCESSING.

Lowercase Text:

hello, my name is rutva sawarkar. my email addresses are rutva@gmail.com and rutva123@yahoo.com. you can contact me at 9876543210. i am a b.tech cse student learning python, data analysis and machine learning. python is easy to learn and python is very useful for data analysis. regular expressions are useful for email validation and text processing.

Text starts with "Hello": **true**

Text contains "Python": **true**

Number of occurrences of "Python": 3

Conclusions: String functions and regular expressions (Regex) are important features of JavaScript that make text processing easier and more efficient. String functions can be used to search, compare, split, replace, trim, and change text, while Regex helps identify and work with specific patterns in the given data. These techniques can be applied to email validation, data extraction, keyword searching, and basic text analysis. When combined with HTML for creating the webpage and CSS for styling, JavaScript provides the required functionality to make the application interactive and user-friendly. Overall, learning string functions and Regex provides a strong foundation for handling text-based data and developing practical web applications.



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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study No.: 06

1. **Case Study Title:** To perform basic string operations ; to reverse a string and count number of vowels in a paragraph.
2. **Case Study Program Code:**

```
<!DOCTYPE html>
<html>
<head>
  <title>String Operations</title>

  <style>
    body {
      font-family: Arial;
      background-color: #f2f2f2;
      text-align: center;
      padding: 30px;
    }

    .box {
      background: white;
      width: 550px;
      margin: auto;
      padding: 25px;
      border-radius: 10px;
    }

    input, textarea {
      width: 90%;
      padding: 10px;
      margin: 10px;
    }

    textarea {
      height: 100px;
    }

    button {
      padding: 10px 20px;
      margin: 5px;
      background-color: black;
      color: white;
      border: none;
    }

    .result {
      font-weight: bold;
      margin-top: 15px;
    }
  </style>
</head>
```

```
<body>
  <div class="box">

    <h1>String Operations</h1>

    <!-- Reverse String -->

    <h2>1. Reverse a String</h2>

    <input type="text" id="string"
      | placeholder="Enter a string">

    <br>

    <button onclick="reverseString()">Reverse</button>

    <p id="reverseResult" class="result"></p>

    <!-- Count Vowels -->

    <h2>2. Count Vowels</h2>

    <textarea id="paragraph"
      | placeholder="Enter a paragraph"></textarea>

    <br>

    <button onclick="countVowels()">Count Vowels</button>

    <p id="vowelResult" class="result"></p>

  </div>

  <script>

    // Reverse a string
    function reverseString() {

      let str = document.getElementById("string").value;

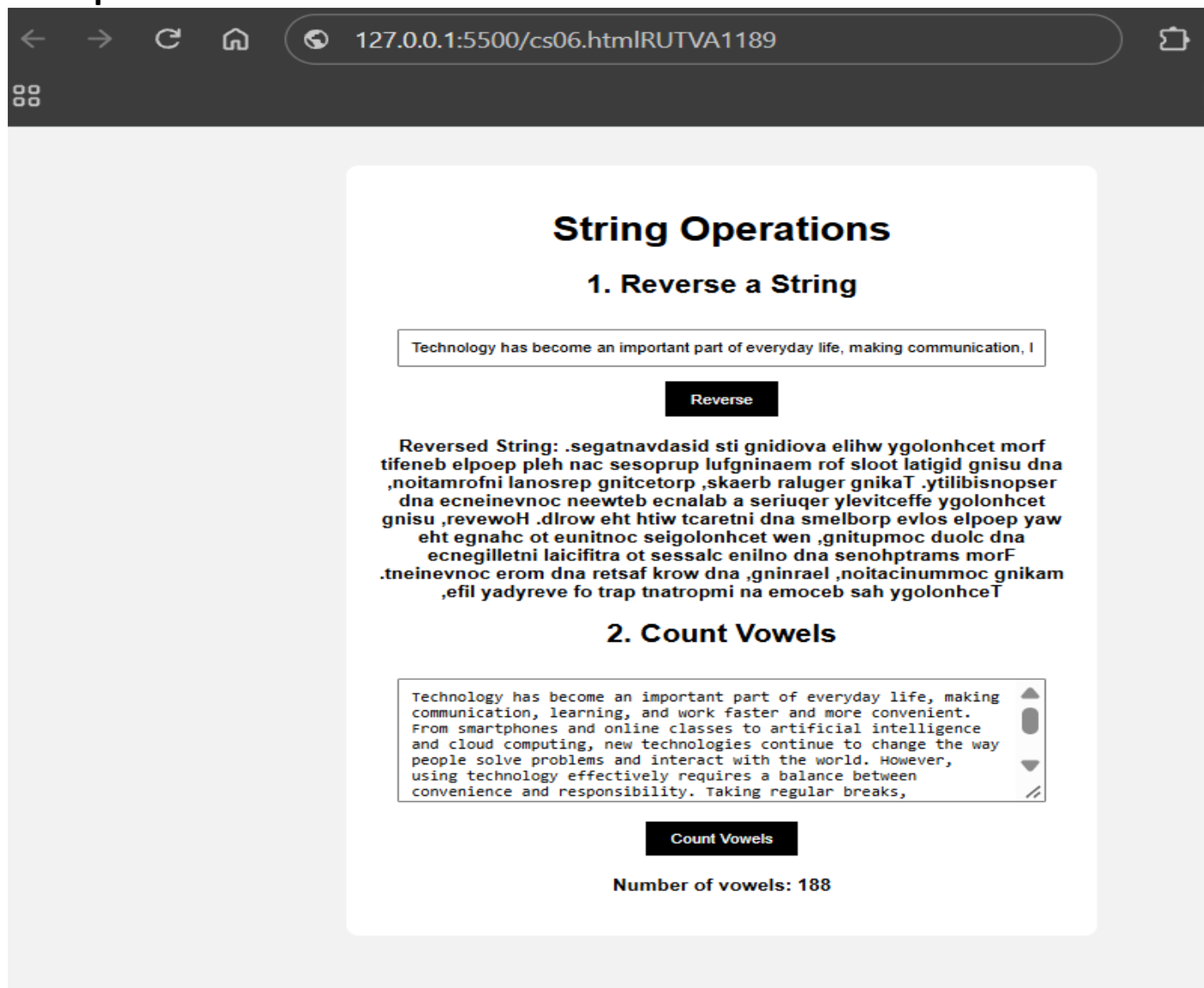
      let reversed = str.split("").reverse().join("");
```

```

91     document.getElementById("reverseResult").innerHTML =
92         "Reversed String: " + reversed;
93 }
94
95
96 // Count vowels
97 function countVowels() {
98
99     let text = document.getElementById("paragraph").value;
100
101     let count = 0;
102
103     for (let i = 0; i < text.length; i++) {
104
105         let ch = text[i].toLowerCase();
106
107         if (ch == "a" ||
108             ch == "e" ||
109             ch == "i" ||
110             ch == "o" ||
111             ch == "u") {
112
113             count++;
114         }
115     }
116
117     document.getElementById("vowelResult").innerHTML =
118         "Number of vowels: " + count;
119 }
120
121 </script>
122
123 </body>
124 </html>

```

3.Output:



4.Result/Conclusion: The JavaScript string operations such as substring, indexOf, split, and replace were successfully performed on the given string. The practical helped in understanding how JavaScript string methods can be used to extract, search, split, and replace text efficiently.



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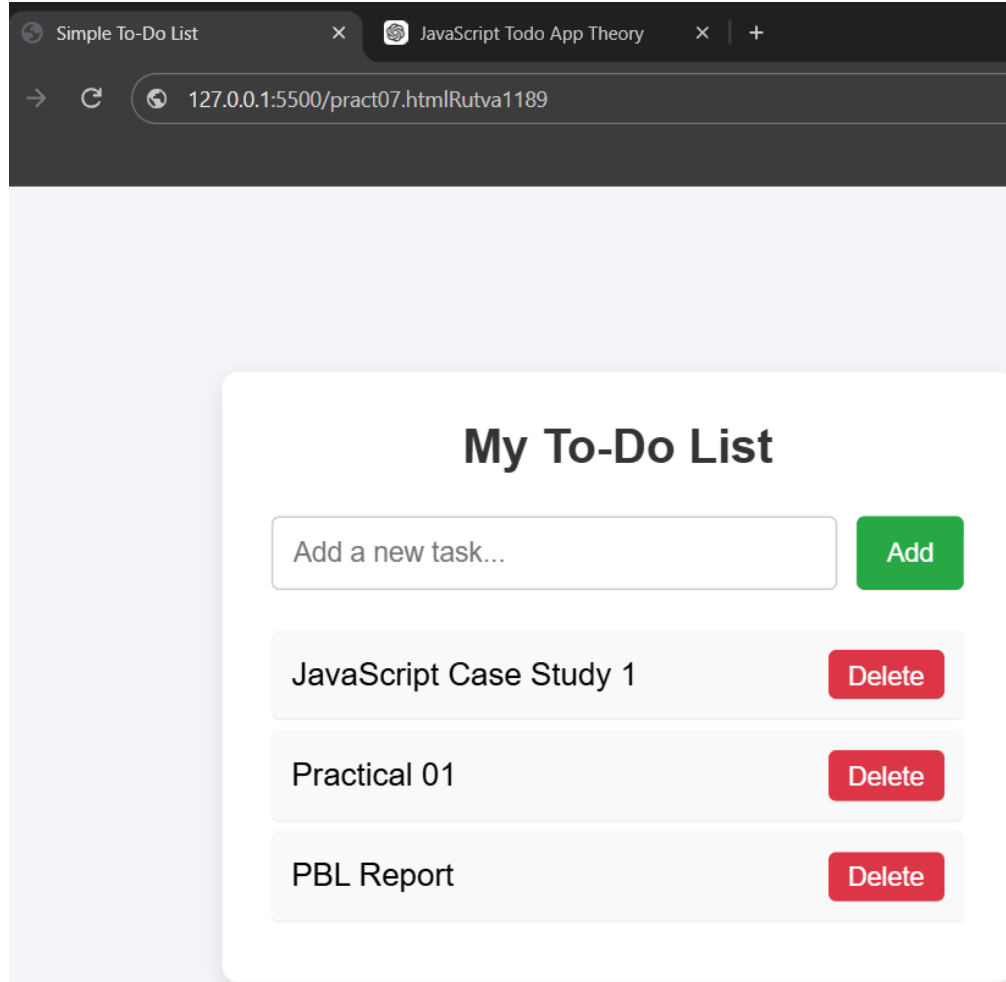
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Experiment No.: 7

- 1. Title:** Perform DOM traversal and updates; develop a to-do list app with add, edit, and delete features.
- 2. Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.
- 3. Theory:** DOM (Document Object Model) traversal is the process of accessing and moving through HTML elements using JavaScript. Methods such as `getElementById()`, `querySelector()`, `parentElement`, `children`, and `nextElementSibling` are used to locate and navigate elements in the DOM. DOM updates allow JavaScript to dynamically change webpage content, attributes, and styles without reloading the page. A **To-Do List application** demonstrates these concepts by allowing users to add new tasks, edit existing tasks, and delete completed or unwanted tasks. JavaScript creates and modifies HTML elements using methods such as `createElement()`, `appendChild()`, `textContent`, and `remove()`. Event listeners such as `click` are used to perform actions when the user interacts with buttons. Thus, DOM traversal and updates make the application interactive and dynamic.



Conclusions: The To-Do List application successfully demonstrates the use of **JavaScript DOM traversal and manipulation**. Tasks can be dynamically added, edited, and deleted based on user actions. This experiment provides a basic understanding of how JavaScript interacts with HTML elements and how event handling can be used to create interactive web applications.



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Case Study No.: 07

1. Case Study Title: To Demonstrate the Events Handling in JavaScript

- Access and manipulate form elements using JavaScript.
- Validate user input (e.g., email, empty fields).
- Use various form events (submit, change, focus). Create the following HTML form and use required JavaScript and events in a given form.

2. Case Study Program Code:

```
cs07.html > html
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Event Handling Form</title>
5
6      <style>
7          body {
8              font-family: Arial, sans-serif;
9              background-color: #f2f2f2;
10             padding: 30px;
11         }
12
13         .container {
14             width: 450px;
15             margin: auto;
16             background-color: white;
17             padding: 25px;
18             border-radius: 10px;
19         }
20
21         h1 {
22             text-align: center;
23         }
24
25         label {
26             display: block;
27             margin-top: 12px;
28             font-weight: bold;
29         }
30
31         input, select {
32             width: 95%;
33             padding: 9px;
34             margin-top: 5px;
35         }
36
37         button {
38             margin-top: 20px;
39             padding: 10px 25px;
40             background-color: black;
41             color: white;
42             border: none;
43             cursor: pointer;
44         }
45     </style>
46 </head>
47 <body>
48     <div class="container">
49         <h1>Student Registration Form</h1>
50
51         <form id="myForm">
52             <label>Full Name</label>
53             <input type="text" id="name" placeholder="Enter your name">
54             <p id="nameError" class="error"></p>
55
56             <label>Email</label>
57             <input type="text" id="email" placeholder="Enter your email">
58             <p id="emailError" class="error"></p>
59
60             <label>Course</label>
61             <select id="course">
62                 <option value=""-- Select Course --</option>
63                 <option value="CSE">Computer Science</option>
64                 <option value="DS">Data Science</option>
65                 <option value="IT">Information Technology</option>
66             </select>
67
68             <label>Phone Number</label>
69             <input type="text" id="phone" placeholder="Enter 10 digit number">
70             <p id="phoneError" class="error"></p>
71
72             <button type="submit">Register</button>
73         </form>
74     </div>
75 </body>
76 </html>
```

```
46     button:hover {
47         background-color: #555;
48     }
49
50     .error {
51         color: red;
52         font-size: 13px;
53     }
54
55     #message {
56         font-weight: bold;
57         margin-top: 15px;
58     }
59 </style>
60 </head>
61
62 <body>
63
64     <div class="container">
65
66         <h1>Student Registration Form</h1>
67
68         <form id="myForm">
69
70             <label>Full Name</label>
71             <input type="text" id="name" placeholder="Enter your name">
72             <p id="nameError" class="error"></p>
73
74             <label>Email</label>
75             <input type="text" id="email" placeholder="Enter your email">
76             <p id="emailError" class="error"></p>
77
78             <label>Course</label>
79             <select id="course">
80                 <option value=""-- Select Course --</option>
81                 <option value="CSE">Computer Science</option>
82                 <option value="DS">Data Science</option>
83                 <option value="IT">Information Technology</option>
84             </select>
85
86             <label>Phone Number</label>
87             <input type="text" id="phone" placeholder="Enter 10 digit number">
88             <p id="phoneError" class="error"></p>
```

```

88     <p id="phoneError" class="error"></p>
89
90     <button type="submit">Submit</button>
91
92     <p id="message"></p>
93
94 </form>
95
96 </div>
97
98
99 <script>
100     // Access form elements
101     let form = document.getElementById("myForm");
102     let name = document.getElementById("name");
103     let email = document.getElementById("email");
104     let course = document.getElementById("course");
105     let phone = document.getElementById("phone");
106
107
108     // Focus event
109     name.addEventListener("focus", function() {
110         name.style.backgroundColor = "#eeeeee";
111     });
112
113
114     // Change event
115     course.addEventListener("change", function() {
116         document.getElementById("message").innerHTML =
117             "Selected Course: " + course.value;
118     });
119
120
121     // Email validation using input event
122     email.addEventListener("input", function() {
123         let pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
124
125         if (!pattern.test(email.value)) {
126

```

```

129             document.getElementById("emailError").innerHTML =
130                 "Enter a valid email.";
131         }
132     } else {
133         document.getElementById("emailError").innerHTML = "";
134     }
135 });
136
137
138 // Phone validation using input event
139 phone.addEventListener("input", function() {
140
141     let pattern = /^[0-9]{10}$/;
142
143     if (!pattern.test(phone.value)) {
144         document.getElementById("phoneError").innerHTML =
145             "Phone number must contain 10 digits.";
146     }
147     else {
148         document.getElementById("phoneError").innerHTML = "";
149     }
150 });
151
152
153 // Submit event
154 form.addEventListener("submit", function(event) {
155
156     // Prevent actual form submission
157     event.preventDefault();
158
159     let valid = true;
160
161     // Name validation
162     if (name.value.trim() === "") {
163
164         document.getElementById("nameError").innerHTML =
165             "Name cannot be empty.";
166         valid = false;
167     }
168

```

```

182         valid = false;
183     }
184
185
186     // Course validation
187     if (course.value === "") {
188
189         document.getElementById("message").innerHTML =
190             "Please select a course.";
191         valid = false;
192     }
193
194
195     // Phone validation
196     let phonePattern = /^[0-9]{10}$/;
197
198     if (!phonePattern.test(phone.value)) {
199
200         document.getElementById("phoneError").innerHTML =
201             "Enter a valid 10-digit phone number.";
202         valid = false;
203     }
204
205
206     // Final result
207     if (valid) {
208
209         document.getElementById("message").innerHTML =
210             "Form submitted successfully!";
211     }
212
213
214 });
215
216
217 </script>
218
219 </body>
220 </html>

```

4. Output:

Student Registration Form

Full Name
Rutva Sawarkar

Email
rutva@gmail.com

Course
Computer Science

Phone Number
9307367867

Submit

Form submitted successfully!

Student Registration Form

Full Name
Rutva Sawarkar

Email
rutva gmail.com
Enter a valid email.

Course
-- Select Course --

Phone Number
11
Enter a valid 10-digit phone number.

Submit

Please select a course.

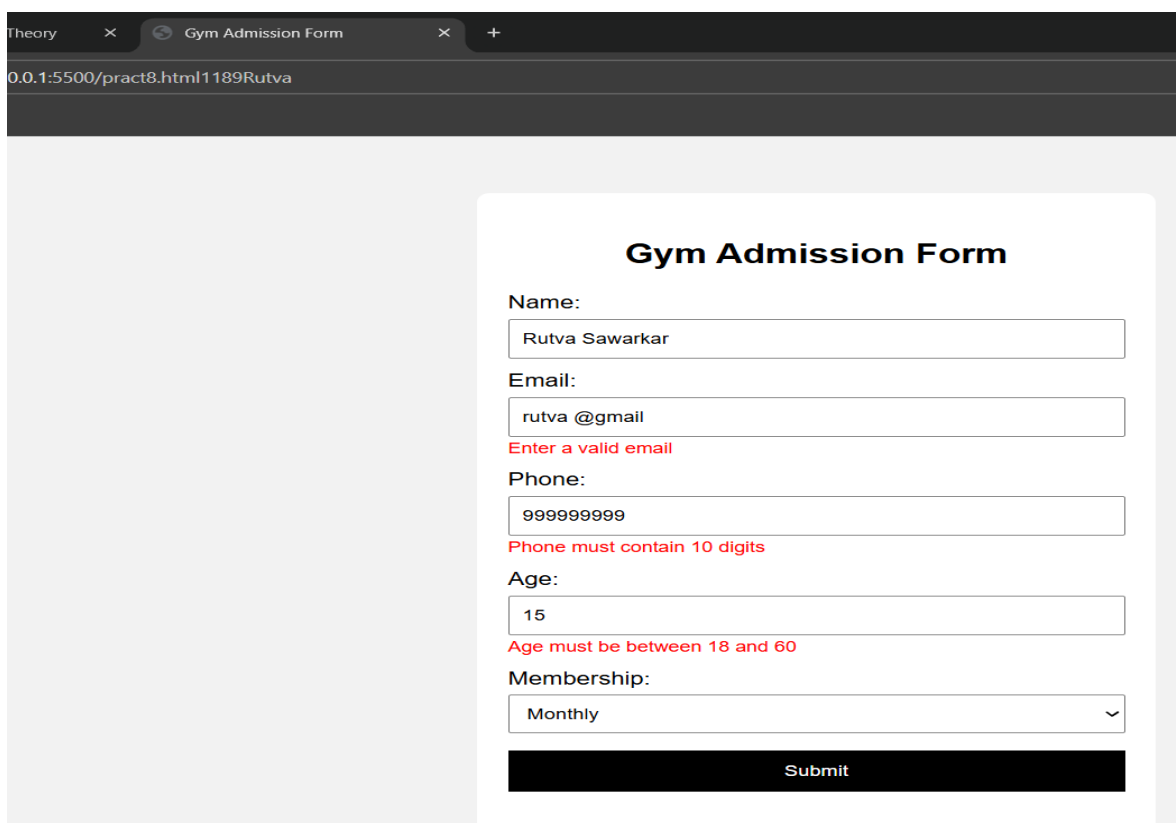
4. Result / Conclusion: The HTML form was successfully created using JavaScript event handling. The form elements were accessed and validated using JavaScript, and focus, change, and submit events were successfully implemented.

Experiment No.: 8

- 1. Title:** Access and validate form fields using events; design a gym admission form with live input checks.
- 2. Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. Theory: Form validation in JavaScript is used to check whether the data entered by a user is correct and complete before submitting a form. JavaScript can access form fields using DOM methods such as `getElementById()` and respond to user actions using events like `input`, `change`, and `submit`. In a **Gym Admission Form**, live input validation can be used to check fields such as name, email, phone number, age, and membership type while the user is entering data. If the entered data is invalid, an error message can be displayed immediately using DOM manipulation. Regular expressions can also be used to validate formats such as email addresses and phone numbers. This makes the form more user-friendly and prevents incorrect data from being submitted.

4. Output :



The screenshot shows a web browser window with the title 'Gym Admission Form'. The URL bar displays '0.0.1:5500/pract8.html1189Rutva'. The form itself is titled 'Gym Admission Form' and contains the following fields and validation messages:

- Name:** Rutva Sawarkar
- Email:** rutva@gmail.com
Enter a valid email
- Phone:** 9999999999
Phone must contain 10 digits
- Age:** 15
Age must be between 18 and 60
- Membership:** Monthly (dropdown menu)
- Submit** button

The screenshot shows a web browser window with the title 'Gym Admission Form'. The address bar displays '/pract8.html1189Rutva'. The main content area features a white form box with the title 'Gym Admission Form' in bold. The form contains the following fields and values:

- Name: Rutva Sawarkar
- Email: rutva@gmail.com
- Phone: 9999999999
- Age: 18
- Membership: Quarterly (dropdown menu)

Below the form fields is a black 'Submit' button. Underneath the button, a green message states 'Gym admission submitted successfully!'.

Conclusions: The Gym Admission Form successfully demonstrates how JavaScript events and DOM methods can be used to access and validate form fields. Live input checks provide immediate feedback to the user and ensure that valid information is submitted. This experiment helps in understanding event handling, form validation, DOM manipulation, and basic JavaScript interaction with HTML forms.



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CASE STUDY 08

Case Study: Create an Online Job Application Form Using Javascript.

PROGRAM CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Online Job Application Form</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
      margin: 0;
      padding: 20px;
    }

    .container {
      width: 70%;
      margin: auto;
      background: white;
      padding: 25px;
      border-radius: 8px;
      box-shadow: 0 0 10px #aaa;
    }

    h1 {
      text-align: center;
      color: #333;
    }

    h2 {
      background-color: #ddd;
      padding: 8px;
      font-size: 18px;
    }

    label {
      display: block;
      margin-top: 10px;
      font-weight: bold;
    }

    input, select, textarea {
      width: 95%;
      padding: 9px;
      margin-top: 5px;
      border: 1px solid #aaa;
      border-radius: 4px;
    }

    textarea {
      height: 70px;
    }

    .gender {
      width: auto;
    }

    .error {
      color: red;
    }

    .success {
      color: green;
      text-align: center;
      font-weight: bold;
    }

    button {
      background-color: #333;
      color: white;
      padding: 12px 25px;
      border: none;
      border-radius: 5px;
      cursor: pointer;
      margin-top: 20px;
    }

    button:hover {
      background-color: #555;
    }
  </style>
</head>

<body>
  <div class="container">

    <h1>Online Job Application Form</h1>

    <form id="jobForm">

      <!-- Personal Information -->
      <h2>Personal Information</h2>

      <label>Full Name</label>
      <input type="text" id="name" placeholder="Enter your full name">
      <div id="nameError" class="error"></div>

      <label>Date of Birth</label>
      <input type="date" id="dob">
      <div id="dobError" class="error"></div>

      <label>Gender</label>
      <input type="radio" class="gender" name="gender" value="Male"> Male
      <input type="radio" class="gender" name="gender" value="Female"> Female
      <input type="radio" class="gender" name="gender" value="Other"> Other
      <div id="genderError" class="error"></div>

      <label>Email</label>
      <input type="text" id="email" placeholder="example@gmail.com">

    </form>

  </div>
</body>
</html>
```

```

110 <div id="emailError" class="error"></div>
111
112 <label>Mobile Number</label>
113 <input type="text" id="mobile" placeholder="10-digit mobile number">
114 <div id="mobileError" class="error"></div>
115
116 <label>Address</label>
117 <textarea id="address" placeholder="Enter your address"></textarea>
118 <div id="addressError" class="error"></div>
119
120 <label>City</label>
121 <input type="text" id="city">
122
123 <label>State</label>
124 <input type="text" id="state">
125
126 <label>PIN Code</label>
127 <input type="text" id="pin">
128 <div id="pinError" class="error"></div>
129
130
131 <!-- Educational Qualification -->
132 <h2>Educational Qualification</h2>
133
134 <label>Highest Qualification</label>
135 <select id="qualification">
136   <option value=""-- Select Qualification --</option>
137   <option>B.Tech</option>
138   <option>B.E.</option>
139   <option>B.Sc</option>
140   <option>BCA</option>
141   <option>M.Tech</option>
142   <option>MCA</option>
143   <option>M.Sc</option>
144   <option>Other</option>
145 </select>
146
147 <label>University / Institute</label>
148 <input type="text" id="university">
149
150 <label>Passing Year</label>
151 <input type="number" id="year" placeholder="YYYY">
152
153 <label>Percentage / CGPA</label>
154 <input type="text" id="percentage" placeholder="Enter percentage or CGPA">
155
156
157 <!-- Professional Information -->
158 <h2>Professional Information</h2>
159
160 <label>Applying For / Job Position</label>
161 <input type="text" id="position">

```

```

163 <label>Total Work Experience</label>
164 <input type="text" id="experience" placeholder="e.g. 2 years">
165
166 <label>Current / Previous Organization</label>
167 <input type="text" id="organization">
168
169 <label>Current Designation</label>
170 <input type="text" id="designation">
171
172 <label>Expected Salary</label>
173 <input type="number" id="salary" placeholder="Enter expected salary">
174
175
176 <!-- Technical Skills -->
177 <h2>Technical Skills</h2>
178
179 <label>Programming Languages</label>
180 <input type="text" id="programming"
181   placeholder="e.g. C, C++, Java, Python">
182
183 <label>Technical Skills</label>
184 <textarea id="skills"
185   placeholder="e.g. HTML, CSS, JavaScript, SQL"></textarea>
186
187 <label>Certifications</label>
188 <textarea id="certifications"
189   placeholder="Enter your certifications"></textarea>
190
191
192 <button type="submit">Submit Application</button>
193
194 <p id="result" class="success"></p>
195
196 </form>
197 </div>
198
199
200 <script>
201
202 // Get form elements
203 const form = document.getElementById("jobForm");
204
205 const name = document.getElementById("name");
206 const email = document.getElementById("email");
207 const mobile = document.getElementById("mobile");
208 const pin = document.getElementById("pin");
209 const dob = document.getElementById("dob");
210 const address = document.getElementById("address");
211
212 // Email validation
213 email.addEventListener("input", function () {
214
215   let pattern = /^[^s@]+@[^s@]+\.[^s@]+$/;

```

```

217   if (!pattern.test(email.value)) {
218     document.getElementById("emailError").innerHTML =
219       "Enter a valid email address.";
220   } else {
221     document.getElementById("emailError").innerHTML = "";
222   }
223 });
224
225
226 // Mobile validation
227 mobile.addEventListener("input", function () {
228
229   let pattern = /^[0-9]{10}$/;
230
231   if (!pattern.test(mobile.value)) {
232     document.getElementById("mobileError").innerHTML =
233       "Mobile number must contain exactly 10 digits.";
234   } else {
235     document.getElementById("mobileError").innerHTML = "";
236   }
237 });
238
239
240 // Name validation
241 name.addEventListener("input", function () {
242
243   if (name.value.trim().length < 3) {
244     document.getElementById("nameError").innerHTML =
245       "Name must contain at least 3 characters.";
246   } else {
247     document.getElementById("nameError").innerHTML = "";
248   }
249 });
250
251
252 // PIN validation
253 pin.addEventListener("input", function () {
254
255   let pattern = /^[0-9]{6}$/;
256
257   if (!pattern.test(pin.value)) {
258     document.getElementById("pinError").innerHTML =
259       "PIN code must contain exactly 6 digits.";
260   } else {
261     document.getElementById("pinError").innerHTML = "";
262   }
263 });

```

```

329 let gender = document.querySelector(
330   'input[name="gender"]:checked'
331 );
332
333 if (!gender) {
334   document.getElementById("genderError").innerHTML =
335     "Please select your gender.";
336   valid = false;
337 } else {
338   document.getElementById("genderError").innerHTML = "";
339 }
340
341 // Address
342 if (address.value.trim().length < 10) {
343   document.getElementById("addressError").innerHTML =
344     "Please enter a valid address.";
345   valid = false;
346 }
347
348 // Final result
349 if (valid) {
350
351   document.getElementById("result").innerHTML =
352     "Application submitted successfully!";
353
354   form.reset();
355
356 } else {
357
358   document.getElementById("result").innerHTML =
359     "Please correct the errors before submitting.";
360 }
361
362 });
363
364 </script>
365
366 </body>
367 </html>

```

3. Output:

127.0.0.1:5500/cs8.htmlRUTVA1189

Online Job Application Form

Personal Information

Full Name

R

Name must contain at least 3 characters.

Date of Birth

06 - 09 - 2026

Gender

☐ Male ☒ Female ☐ Other

Email

12r@.com

Enter a valid email address.

Mobile Number

1111

Mobile number must contain exactly 10 digits.

Address

-

Please enter a valid address.

City

NAGPUR

State

Maharashtra

PIN Code

440008

Educational Qualification

Highest Qualification

B. Tech

University / Institute

Symbiosis Institute of Technology

Passing Year

2011

Percentage / CGPA

7.0

Professional Information

Applying For / Job Position

Data Analyst

Total Work Experience

1

Current / Previous Organization

XYZ

Current Designation

Intern

Expected Salary

999987

127.0.0.1:5500/cs8.htmlRUTVA1189

Online Job Application Form

Personal Information

Full Name

Rutva

Date of Birth

06 - 09 - 2026

Gender

☐ Male ☒ Female ☐ Other

Email

rutva@gmail.com

Mobile Number

1111111111

Address

XYZ, Nagpur

City

NAGPUR

State

Maharashtra

Technical Skills

Programming Languages

C, C++, JAVA, Python

Technical Skills

JS, SQL, CSS, HTML

Certifications

Redhat, Google, Springboard

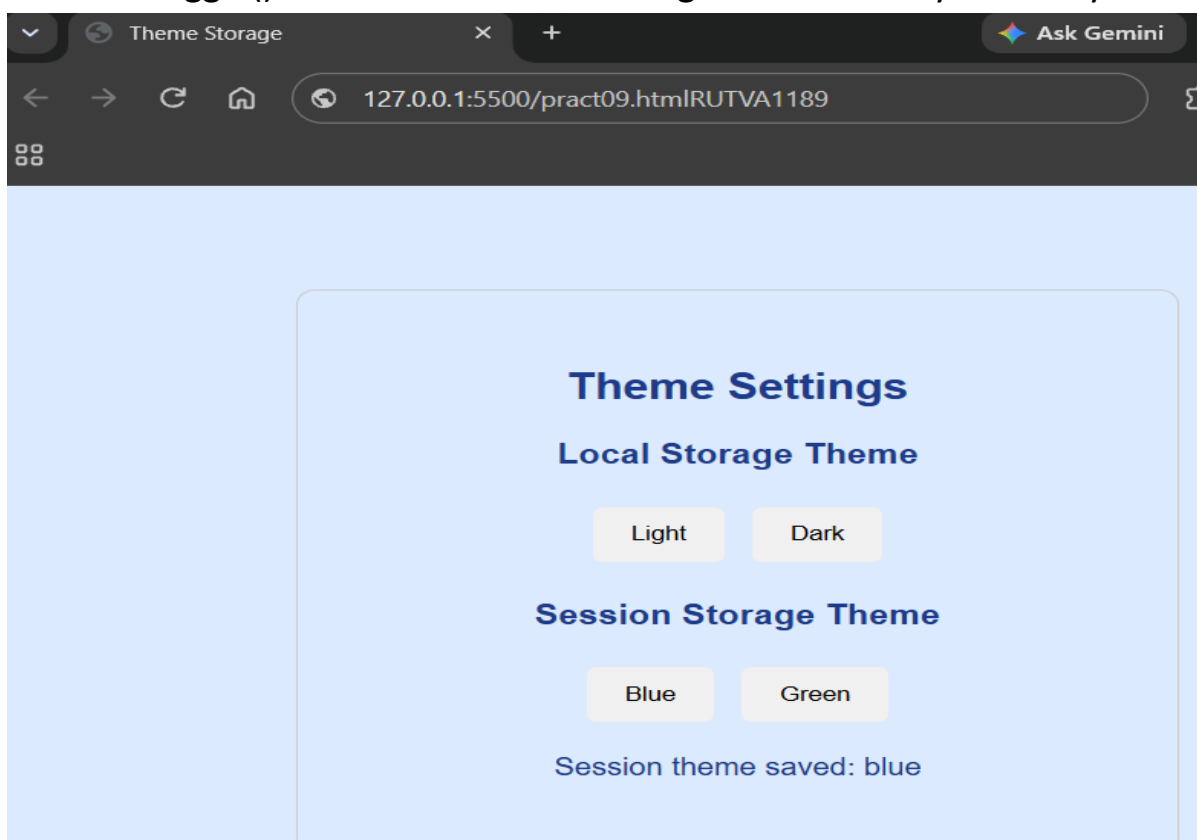
Submit Application

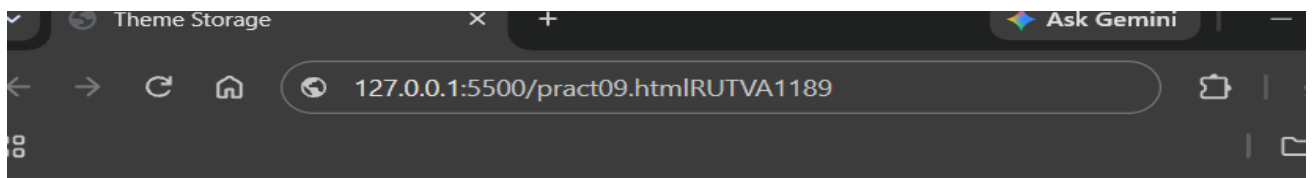
Please correct the errors before submitting.

4. Result / Conclusion: The Online Job Application Form was successfully created using HTML, CSS, and JavaScript. JavaScript was used to validate the applicant's personal, contact, and educational details before submission. The form checks whether the entered information is valid and displays appropriate messages for incorrect or missing details. Thus, the case study demonstrates how JavaScript can be used for form validation, error handling, and creating an interactive webbased application form.

Experiment No.: 9

- 1. Title:** Implementation of localStorage and sessionStorage to Save User Preferences
- 2. Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.
- 3. Theory:** Web Storage is a feature of JavaScript that allows websites to store data in the user's browser. It provides two main storage methods: localStorage and sessionStorage. localStorage is used to store data that remains available even after closing and reopening the browser. In this practical, localStorage is used to save the user's theme preference, such as light mode or dark mode. sessionStorage is used to store data for a particular browser session. In this practical, it is used to save the user's name. The stored data remains available while the session continues and is generally removed when the browser tab or session ends. JavaScript methods such as setItem() and getItem() are used to save and retrieve data from storage. The classList.toggle() method is used to change the theme dynamically.





Theme Settings

Local Storage Theme

Light

Dark

Session Storage Theme

Blue

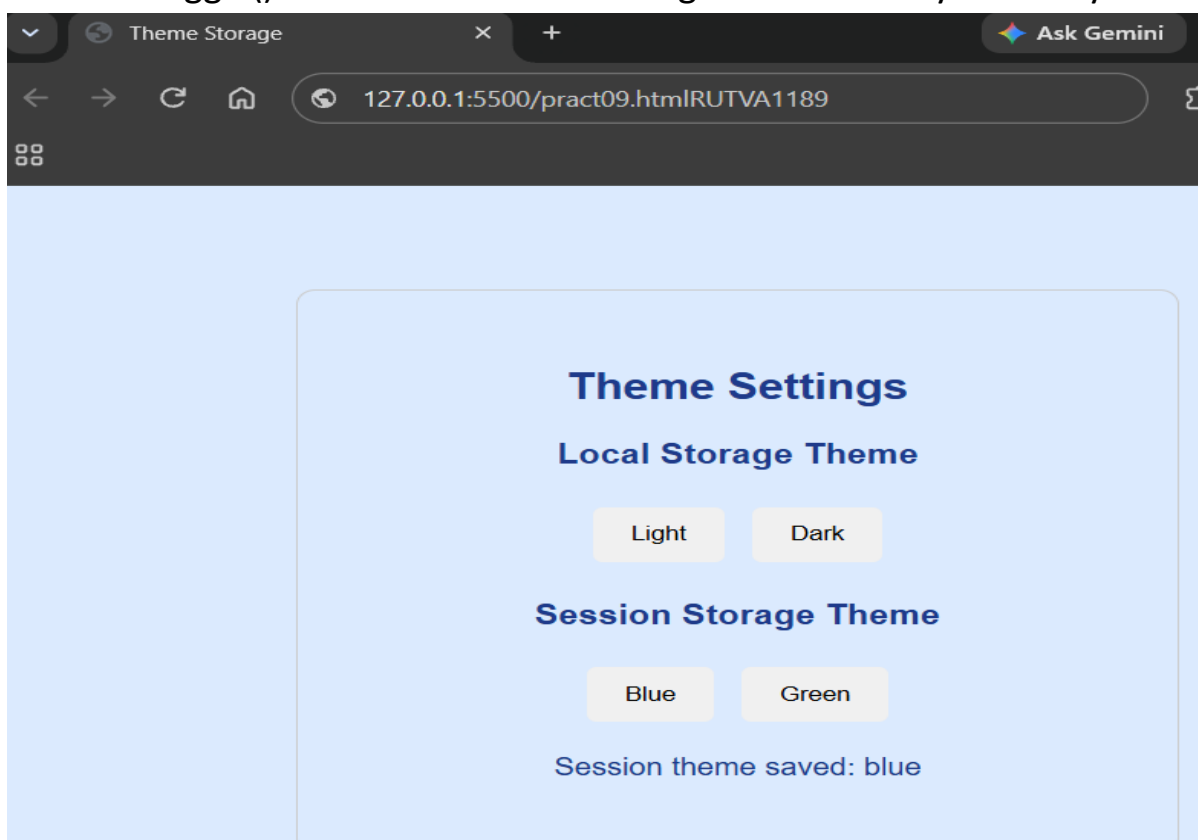
Green

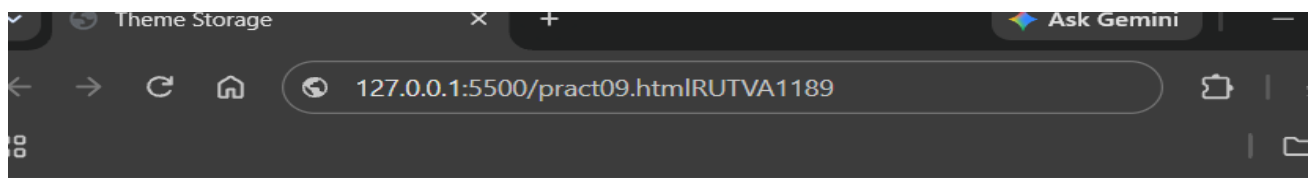
Local theme saved: light

4. Conclusions: In this practical, we successfully implemented `localStorage` and `sessionStorage` to save user preferences. `localStorage` was used to store the theme settings, while `sessionStorage` was used to store the username. The theme was changed dynamically using JavaScript, and stored preferences were retrieved when the page was reloaded. This practical helped us understand browser storage and its use in web development.

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Theme Settings

Local Storage Theme

Light

Dark

Session Storage Theme

Blue

Green

Local theme saved: light

4. Conclusions: In this practical, we successfully implemented `localStorage` and `sessionStorage` to save user preferences. `localStorage` was used to store the theme settings, while `sessionStorage` was used to store the username. The theme was changed dynamically using JavaScript, and stored preferences were retrieved when the page was reloaded. This practical helped us understand browser storage and its use in web development.